

Ordered pair (Kuratowski, 1921)

$$(x, y) \neq (y, x) \\ \text{if } x \neq y$$

$$(x, y) = \{\{x\}, \{x, y\}\} \\ (y, x) = \{\{y\}, \{y, x\}\}$$

Proposition  $(x_1, y_1) = (x_2, y_2) \Leftrightarrow x_1 = x_2 \text{ and } y_1 = y_2$

$$\Rightarrow (x_1, y_1) = (x_2, y_2) \Rightarrow \{\{x_1\}, \{x_1, y_1\}\} = \{\{x_2\}, \{x_2, y_2\}\}$$

$$\textcircled{1} \quad x_1 = y_1 \\ \Downarrow$$

$$\{x_1, y_1\} = \{x_1\} \Rightarrow (x_1, y_1) = \{\{x_1\}, \{x_1\}\} \Rightarrow x_2 = y_2 \\ \Rightarrow x_1 = x_2$$

$$\textcircled{2} \quad x_1 \neq y_1 \text{ and } x_2 \neq y_2 \Rightarrow \{x_1\} \neq \{x_1, y_1\}$$

$$\text{and } \{x_2\} \neq \{x_2, y_2\}$$

$$\{\{x_1\}, \{x_1, y_1\}\} = \{\{x_2\}, \{x_2, y_2\}\} \quad \text{if } \{x_1\} = \{x_2\} \quad \Leftarrow$$

$$\Rightarrow x_1 = x_2 \Rightarrow y_1 = y_2$$

Def The Cartesian product of sets  $X$  and  $Y$  is the set  $X \times Y = \{(x, y) \mid x \in X, y \in Y\}$

$X \times Y = Y \times X$  doesn't hold in general

Def A relation  $R$  on the set  $X$  is a subset of  $X \times X$

Example  $X = \{1, 2, 3\}$   $X \times X = \{(1, 1), (1, 2), (1, 3), \dots, (3, 1), (3, 2), (3, 3)\}$

$$R_1 = \{(1, 1), (2, 2), (3, 3)\}$$

$$R_2 = \{(1, 2), (1, 3), (2, 3)\}$$

$$R_3 = \{(1, 2), (2, 1), (1, 3), (3, 1), (2, 3), (3, 2)\}$$

$$R_4 = \{(1, 1)\}$$

$$R_5 = \emptyset$$

$$R_6 = X \times X$$

Def Let  $X \neq \emptyset$ . A partial order on  $X$  is a relation  $R \subseteq X \times X$  s.t.

1.  $R$  is reflexive,  $(x, x) \in R \quad \forall x \in X$

2.  $R$  is anti-symmetric.

if  $(x, y) \in R$  and  $(y, x) \in R \Rightarrow x = y$

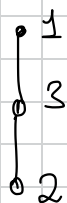
3.  $R$  is transitive

if  $(x, y) \in R$  and  $(y, z) \in R \Rightarrow (x, z) \in R$

Example  $x \in \{1, 2, 3\}$

$$\textcircled{1} \quad 1 \preceq 1, 2 \preceq 2, 3 \preceq 3, 1 \preceq 2, 1 \preceq 3, 2 \preceq 3$$

$$\textcircled{2} \quad (1, 1), (2, 2), (3, 3), (1, 2), (1, 3), (3, 2)$$



## Natural numbers

$$\begin{array}{ccccccc} \emptyset & \{\emptyset\} & \{\emptyset, \{\emptyset\}\} & \{\emptyset, \{\emptyset, \{\emptyset\}\}\} & \dots & N_{n+1} = \{N_n\} \cup \{\emptyset\} \\ \parallel & N_1 & N_2 & N_3 & & \\ N_0 & & & & & \end{array}$$

Def Let  $N$  be a set with a distinguished element denoted as  $0$  and a function  $s: N \rightarrow N$ . We say that  $(N, s, 0)$  satisfy Peano axioms if

1. There is no  $n \in N$  with  $s(n) = 0$
2. The map  $s$  is injective: if  $s(m) = s(n)$  for some  $m, n \in N$ , then  $m = n$
3. If  $K \subseteq N$  is such that  $0 \in K$  and if  $n \in K \Rightarrow s(n) \in K$ , then  $K = N$

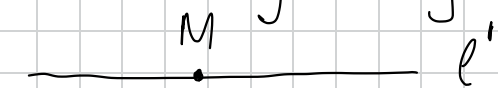
Lecture 2

09.09

Alternative math

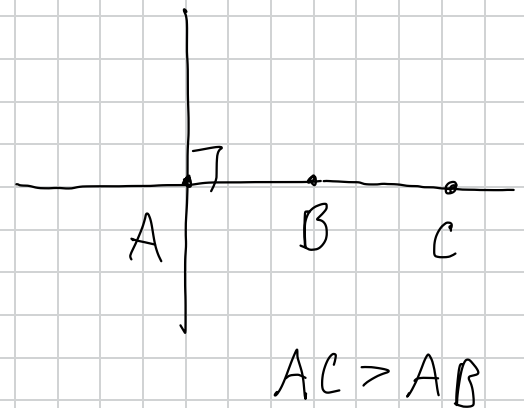
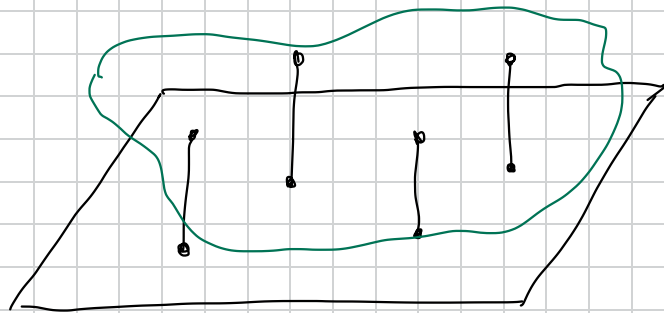
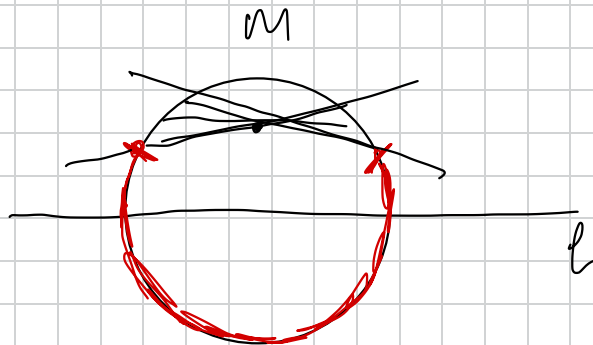
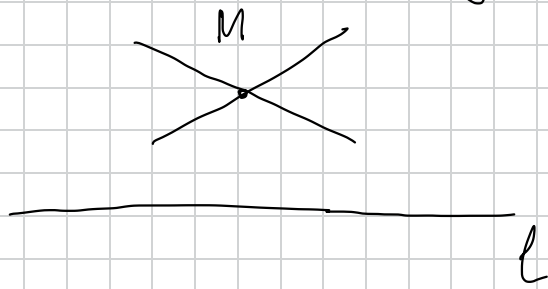
$x \in X$  with probability  $p$

Euclidean geometry

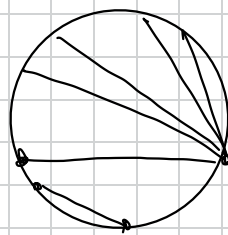
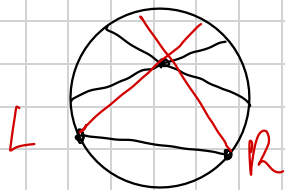


$$\exists ! l' : M \in l', l' \cap l = \emptyset$$

# Hyperbolic geometry



Model:



class of parallel lines

$K$  is inductive if  $0 \in K$  and  $(n \in K) \Rightarrow (S(n) \in K)$

$K \subseteq \mathbb{N}$ . 3rd Peano axiom: if  $X$  is inductive, then  $K = \mathbb{N}$

## Proposition

Every non-zero natural number is a successor

$$\forall n \in \mathbb{N}: n=0 \text{ or } \exists m: S(m)=n$$

Construct  $K = \{n \in \mathbb{N} \mid n=0 \text{ or } \exists m \in \mathbb{N}: S(m)=n\}$

$$0 \in K \quad \text{True}$$

assume  $n \in K$ , prove that  $S(n) \in K$

$$S(n) \rightarrow n \in \mathbb{N}, S(n) = S(m), m=n$$

Then  $K$  is inductive

$$n = n+1$$

$$n+1 = n+2$$

$$n=1 \quad \text{True}$$

$$n=k \quad \text{True}$$



# Arithmetic

## Definition

Let  $X$  be a set. A binary operation on  $X$  is a map

$$\odot: X \times X \rightarrow X$$

$$x \odot y = z$$

$$\odot(x, y) = z$$

## Example

$$X = \{a, b\}$$

$$\odot(a, a) = a \quad \odot(a, b) = b \quad \odot(b, a) = b \quad \odot(b, b) = a$$

## Addition of natural numbers

Def

For all  $m, n \in \mathbb{N}$  define

$$1. \quad m + 0 = m$$

$$2. \quad m + S(n) = S(m + n)$$

$$1 = S(0)$$

$$2 = S(S(0))$$

$$\begin{aligned} 1 + 2 &= 1 + S(1) = S(1 + 1) = S(1 + S(0)) = S(S(1 + 0)) = \\ &= S(S(1)) = S(S(S(0))) = 3 \end{aligned}$$

## Proposition

For all  $m, n, p \in \mathbb{N}$  the following holds:

1.  $m + (n + p) = (m + n) + p$  (associativity)
2.  $0 + n = n$  - neutral element
3. If  $m + p = n + p$  then  $n = m$  - cancellation
4.  $m + n = n + m$  - commutativity

## Proof

$$1. \quad K = \{ p \in \mathbb{N} \mid m + (n + p) = (m + n) + p, \forall m, n \in \mathbb{N} \}$$

$$0 \in K \quad m + (n + 0) = (m + n) + 0 \rightarrow \text{True}$$

$$p \in K \quad m + (n + p) = (m + n) + p$$

Prove that  $m + (n + S(p)) = (m + n) + S(p)$

$$\begin{array}{ccc} \downarrow & & \downarrow \\ m + S(n + p) & & S((m + n) + p) \\ \downarrow & & \parallel \text{ by assumption} \\ S(m + (n + p)) & = & S(m + (n + p)) \end{array}$$



$$4. \quad K = \{ n \in \mathbb{N} \mid \forall m \in \mathbb{N} : m + n = n + m \}$$

$$0 \in K \quad m + 0 = 0 + m \quad \text{True}$$

$$n \in K \quad m + n = n + m$$

$$\begin{aligned} m + S(n) &= S(n) + m \\ \parallel \\ S(m + n) &= S(n + m) \end{aligned}$$

$$K_1 = \{ m \mid \forall n \quad S(n + m) = S(n) + m \}$$

$$0 \in K, \quad S(n+0) = S(n) + 0$$

$$S(n+m) = S(n) + m$$

$$S(n+S(m)) = S(n) + S(m)$$

$$S(S(n+m)) = S(S(n) + m)$$

by assumpt.



### Lecture 3

11.09

Def for all  $m, n \in \mathbb{N}$  define:

1.  $m \cdot 0 = 0$
2.  $m \cdot S(n) = m \cdot n + m$

Example

$$m \cdot 1 = m \cdot S(0) = m \cdot 0 + m = m$$

Proposition

Suppose  $m, n, p \in \mathbb{N}$ . Then

1.  $m \cdot (n+p) = (m \cdot n) + (m \cdot p)$  distributivity
2.  $m \cdot (n \cdot p) = (m \cdot n) \cdot p$  - associativity
3. Neutral element  $1 \cdot m = m \cdot 1 = m$
4.  $m \cdot n = n \cdot m$  commutativity

Proof 1. Induction in  $p$



Base:  $p=0$

$$m \cdot (n+0) \stackrel{?}{=} (m \cdot n) + (m \cdot 0)$$

$$m \cdot n = m \cdot n \quad \text{True}$$

Step: Assume that  $m \cdot (n+p) = (m \cdot n) + (m \cdot p)$

Prove  $m \cdot (n+s(p)) = (m \cdot n) + (m \cdot s(p))$

$$\begin{aligned} \text{LHS: } m \cdot (n+s(p)) &= m \cdot s(n+p) = m \cdot (n+p) + m = \\ &= m \cdot n + m \cdot p + m = m \cdot n + m \cdot s(p) \quad \text{True} \end{aligned}$$



## Ordering $\mathbb{N}$

Def for  $m, n \in \mathbb{N}$  we write  $m \leq n$  if  $\exists c \in \mathbb{N}$   
s.t.  $m+c=n$

## Proposition

For all  $m, n, p \in \mathbb{N}$  we have

1.  $m \leq m$  ( $c=0$ ) (reflexivity)
2. If  $m \leq n$  and  $n \leq m$  then  $m=n$  (antisymmetry)
3. If  $m \leq n$  and  $n \leq p$  then  $m \leq p$
4. If  $m \leq n$  then  $p+m \leq p+n$

## Proof

$$2. \quad m \leq n, n \leq m \quad \exists c: m+c=n \quad \& \quad \exists d: n+d=m$$

$$\begin{aligned} m &= \overset{n}{m+c} + d \\ &\Downarrow \\ c+d &= 0 \end{aligned}$$

$$\textcircled{1} \quad d=0 \Rightarrow c=0$$

$$\textcircled{2} \quad d \neq 0 \Rightarrow \exists d': d = s(d')$$

$$c+d = c+s(d') = s(c+d') = 0 \quad \text{↯}$$

$$\textcircled{3} \quad m \leq n \text{ and } n \leq p \quad \exists c, n = m+c, \exists d: p = n+d \\ \Rightarrow m+(c+d) = (m+c)+d = n+d = p$$

$$\textcircled{4} \quad m \leq n \quad \exists c: m+c = n \quad p+m+c = p+n \\ \Rightarrow p+m \leq p+n$$

Proposition  $\forall m, n \in \mathbb{N}$  we have  $m \leq n$  or  $n \leq m$

Induction  $K = \{n \in \mathbb{N} \mid \forall m \in \mathbb{N} \quad m \leq n \text{ or } n \leq m\}$

$$0 \in K \quad n=0 \quad 0 \leq m, c=m$$

$(n \in K \Rightarrow s(n) \in K)$ :

$$\textcircled{1} \quad m \leq n \Rightarrow \exists c: m+c = n$$

$$s(n) = s(m+c) = m+s(c) \Rightarrow m \leq s(n)$$

$$\textcircled{2} \quad n \leq m$$

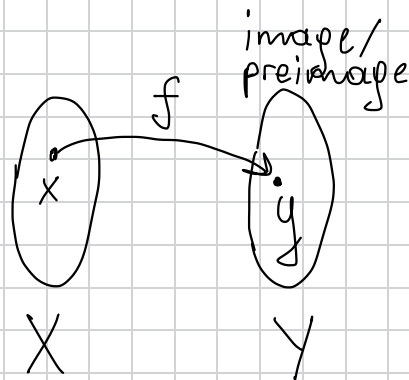
$$\exists c, n+c = m$$

$$\textcircled{a} \quad c \neq 0 \Rightarrow \exists c': c = s(c')$$

$$m = n + s(c') = s(n+c') = s(n) + c'$$

$$\textcircled{b} \quad c=0 \Rightarrow n=m \Rightarrow s(n) = s(m) = m+1 \Rightarrow m \leq s(n)$$

## Sets: finite and infinite



$f$  is injective if  $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$

$f$  is surjective if  $\forall y \in Y \exists x \in X:$   
 $f(x) = y$

$f$  is bijective if it is injective  
and surjective

Def Suppose  $X$  and  $Y$  are sets.

We say that  $X$  and  $Y$  are equipotent or  
have the same cardinality if  $\exists f: X \rightarrow Y$  -  
bijective

def. for  
every  $x \in X$

### Proposition

Suppose  $X, Y, Z$  are sets. Then

1.  $X$  is equipotent with  $X$
2. If  $X$  is equipotent with  $Y$ , then  $Y$  is equipotent with  $X$
3. If  $X \sim Y$  and  $Y \sim Z \Rightarrow X \sim Z$

①  $\text{id}: X \rightarrow X$

②  $\exists f \rightarrow \exists f^{-1}: Y \rightarrow X$  - bijection

③  $\exists f: X \rightarrow Y, \exists g: Y \rightarrow Z \quad h = (g \circ f): X \rightarrow Z$